

CLS Linac Klystron Specifications

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1.0 PURPOSE

This document specifies the requirements for the 2856 MHz (S-band) klystron required for the existing linac modulator and accelerator section. The testing and delivery of the klystron has been discussed in the document.

2.0 GENERAL BACKGROUND

The operating klystrons for the CLS linac have been running 30,000-50,000 hours during the past ten years. New klystrons are needed because the existing ones are nearing end of life and the original manufacturer is no longer in business.

The S-band klystrons provide the operating RF power of 20MW with pulse width 4 μ s at frequency 2856 MHz. The detail requirements are described below.

3.0 DEFINITIONS

CLS	Canadian Light Source Inc.
RF	Radio Frequency
VSWR	Voltage Standing Wave Ratio
pps	Pulse per second
SIP	Sputter Ion Pump
Buyer	Canadian Light Source Inc.
Vendor	Klystron manufacturer
ScandiNova	ScandiNova Systems AB. A modulator manufacturer in Uppsala, Sweden

4.0 TECHNICAL SPECIFICATION REQUIREMENTS

Many of the specifications listed below are driven by the existing specifications on these modulators. which were made by ScandiNova in 2011.

4.1 ELECTRICAL/RF SPECIFICATIONS

• Type	S-band, Pulsed
• Operating frequency	2856 MHz
• Bandwidth	$\geq \pm 2$ MHz
• Operating temperature	20-35 °C
• Peak RF output power	25 MW/max. (20 MW/typical)
• Pulse repetition rate	1-3 pps
• RF pulse length	4.0 μ s
• RF rise/fall time	≈ 1 μ s
• Output VSWR requirement	1.30 : 1
• Efficiency	$\approx 40\%$
• Gain	47-53 dB
• RF drive power	100-400 W
• High voltage	<250 kV ¹
• Beam current	<250 A ¹
• Filament voltage	15 V (reasonable value) ²
• Filament current	15 A (reasonable value) ²
• Waveguide	WR-284 (working in HV vacuum)
• RF output flange	Skarpaas ³
• RF drive input connector	Type "N" female 50 Ω coaxial connector

Note 1: The klystron high voltage and current depend on the existing linac modulator (see section 6) and may be not changeable.

Note 2: The klystron filament voltage and current values depend on the existing linac modulator (see section 6), but they can be changed by replacing a new filament power supply later with ScandiNova. (Not desirable.)

Note 3: Skarpaas waveguide flanges are used in CLS linac waveguide. If the klystron flange is different, vendor shall provide an adapter.

4.2 MECHANICAL SPECIFICATIONS

• Maximum klystron height	1.5 m (see CLS Drawing #0184391 ^[1])
• Weight (klystron and magnet)	< 720 kg (crane restriction)
• Collector cool flow	Shall be approved by CLS
• Body cool flow	Shall be approved by CLS
• Water connector	Shall be approved by CLS
• Modulator sizes	See CLS Drawing #0184391 ^[1]
• Modulator room dimensions	See CLS Drawing #0184392 ^[2]

4.3 VACUUM SPECIFICATIONS

- Vacuum tight leakage <1.0E-10 atm.cc/s of helium
- SIP current $\leq 1\mu\text{A}$ with full RF power
- SIP $\approx 8 - 10 \text{ l/s}$
- SIP controller Supplied with the klystron
- SIP connector/cables Supplied with the klystron

4.4 MAGNET SPECIFICATIONS

- Magnet Permanent magnet preferred ³
- Height Less than 474 mm
- Diameter Provide to and accepted by CLS ²
- Weight <570 kg

Note:

1. The total height of the magnet and the klystron should be less than 2.010 m due to the room limit.
2. The diameter of the magnet should fit to the existing modulators as it shown in the section 8, attachment [2].
3. Electromagnet may be accepted. If vendor supplies electromagnet, the drive voltage, current and cooling requirement must be approved by CLS. Vendor shall supply the drive power supplies.

5.0 MODULATOR INTERFACE

5.1 KLYSTRON INFORMATION TO SCANDINOVA

The existing linac modulators in CLS were made by ScandiNova Systems AB and will be used with the new klystrons. Some modulator parts may need to be modified or replaced to fit the new klystron requirements. Consequently, permission must be given to the CLS to share any relevant klystron specifications with ScandiNova Systems AB for the purposes of costing and modifying the existing CLS modulators.

5.2 ACCEPTANCE TEST

- The klystrons shall be tested in ScandiNova modulators at CLS to measure all parameters in table 1 to meet the requirements in the contract.
- All required test parameters are listed in the table 1.

Table 1 Klystron Parameters in Acceptance Test

Klystron Parameters at Acceptance Tests	Parameter Range
Peak RF output power	25 MW/max.
Pulse repetition rate	1-3 pps
RF pulse length	4.0 μ s
Output VSWR requirement	1.3
Efficiency	$\approx$ 40%
Gain	47-53 dB
RF drive power	100-400 W
High voltage	<250 kV
Beam current	<250 A
Filament voltage	(15) V
Filament current	(15) A

5.3 DRAWINGS AND DOCUMENTATION

All drawings and documentation shall follow CLS Vendor Documentation Specification. [4]

6.0 REFERENCE (LINAC MODULATOR SPECIFICATION)

The linac modulators were manufactured by ScandiNova Systems AB in 2011. The candidate klystrons should operate with the existing modulator specifications. ScandiNova will do limited modification for the new klystrons. E.g. filament power supply and the klystron socket, etc., but shall not require the CLS to use different modulators. The CLS linac modulator specifications are listed in the table 2.

Table 2 ScandiNova Modulator Specification

ScandiNova K2 Modulator Specification		
Parameters	Range	Unit
Peak Power	62.5	MW
Pulse Voltage	250	kV
Pulse Current	250	A
Average Power	0.48	kW
Repetition Rate	1-3	Hz
Pulse Width	4	μs
Voltage Rise Time	1	μs
Voltage Fall Time	1	μs
Voltage Top Flatness	± 0.25	%
Pulse Jitter	≤ ±5	ns
Modulator Efficiency	>80	%
Klystron Filament Voltage	15	V
Klystron Filament Current	15	A

7.0 REFERENCE

- [1] CLS Klystron Height Requirements, CLS Drawing #0184391
- [2] CLS Modulator Room Layout, CLS Drawing #0184392
- [3] CLS 0.7.1.14 Document Control Procedures for Drawings
- [4] CLS 0.4.1.1 Vendor Documentation Specification

REVISION HISTORY

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A	2013-03-22	Original Draft	Xiaofeng Shen
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